

AMRITH S S

B.Tech Information Technology | Data Science • Machine Learning • Software Development

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PROFESSIONAL SUMMARY

B.Tech Information Technology student with hands-on experience building data-driven and full-stack applications using Python, JavaScript, React, Node.js, SQL and machine-learning libraries. Built applications covering symptom-guidance logic, recommendation and ranking systems, authentication, REST APIs, OCR/NLP workflows and NLP text classification. Seeking entry-level opportunities in Data Science, Machine Learning, Data Analytics or Data Engineering.

EDUCATION

Dr. Mahalingam College of Engineering and Technology | B.Tech, Information Technology | Expected 2028 | CGPA: 8.2/10 (up to Semester 5)

TECHNICAL SKILLS

Programming: Python, C, Java, JavaScript

Data & ML: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch

Web & Backend: React, Vite, Tailwind CSS, Node.js, Express, REST APIs, FastAPI, Streamlit

Databases & Security: SQL, SQLite, Prisma, JWT, bcrypt

Tools: Git, Docker, npm

Fundamentals: Data Structures & Algorithms, DBMS, Machine Learning, Computer Networks, Operating Systems

PROJECTS

Rural Health Guide | React, Vite, Express, SQLite, Python, Scikit-learn, PWA

- Developed a low-bandwidth symptom-guidance web application with structured flows for children, adults and older adults.
- Implemented Express REST APIs and a SQLite-backed rule engine using symptoms, duration, severity and red flags to generate guidance.
- Integrated a Python Scikit-learn MLP as an auxiliary early-stage diabetes-risk signal while retaining rule-based emergency guidance.
- Added English/Hindi support, browser voice input, text-to-speech and service-worker caching for low-connectivity use.

What Can I Cook? | React, Tailwind CSS, Express, Prisma, SQLite, JWT

- Developed a full-stack recipe finder that recommends meals from ingredients available to the user.
- Built REST endpoints for ingredient search, recipe details, ranked ingredient matching, favourites and community recipe submissions.
- Implemented ranking using matched ingredients, missing ingredients and match percentage, with category, meal-type, diet and cooking-time filters.
- Added JWT authentication with bcrypt password hashing and optional AI-assisted recipe verification.

Movie Recommender System | Python, Pandas, Scikit-learn, Streamlit

- Built a content-based recommendation pipeline using TF-IDF and CountVectorizer to identify similar movies from metadata.
- Serialized preprocessing and model artifacts with Pickle and integrated recommendations into an interactive Streamlit application.

Cyberbullying Message Detection | Python, NLP, Logistic Regression, Streamlit

- Built a binary NLP classification pipeline that cleans text, compares several models and predicts whether a message is bullying.
- Integrated the model into a simple Streamlit interface for live message checks.

Medicine Strip Text Reader & Dosage Assistant | Python, OCR, NLP, PyTorch, FastAPI

- Designed an OCR-to-NLP workflow to extract medicine-strip text and structure information for a dosage-reminder assistant.
- Worked with PaddleOCR, TrOCR, CRAFT/DBNet, spaCy, Sentence-BERT and LayoutLM across text detection, recognition and NLP components.

CERTIFICATIONS & PROBLEM SOLVING

- HackerRank SQL Certification
- Cisco Networking Academy — Python Essentials
- IBM SkillsBuild — professional and technical learning path
- Solved 130+ Data Structures & Algorithms problems; strengthening C-based DSA and aptitude for placement preparation.

ADDITIONAL

Strengths: Problem solving • Analytical thinking • Continuous learning | **Relocation:** Willing to relocate